

ROUND 2

Build It. Demo It. Defend It.

Problem Statements and Judging Criteria

LYB Mumbai Hackathon 2026

Four problem statements. One working prototype per team.

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These Are Hypothetical Scenarios

All four problem statements describe a fictional company called Plant X.

DISCLAIMER

All problem statements in this document are hypothetical and were written for this hackathon only.

"Plant X" is a fictional company. The scenarios, processes, roles and figures do not describe the actual operations, systems, data or policies of LYB or of any other organization.

Any resemblance to real events, sites or people is coincidental.

WHAT THIS MEANS FOR YOU

Build for Plant X
Not for a real site or system

Create your own sample data
Never use real, personal or confidential data

State your assumptions
Say what you assumed and why

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Choose One. Build It. Prove It Works.

Pick one of four problem statements and build a working web or mobile prototype for it.

THE BRIEF

Code it. Run it live.

This is a software challenge. Your solution is an app, web app or tool that you code and demonstrate working, not slides or a concept.

No data or input is provided, before or during your presentation.

YOUR MISSION

01 Choose
Pick one of the four problem statements and decide which user and problem you will focus on.

02 Build
Code a working prototype that covers the five core expectations. Bonus items are optional.

03 Defend
Demo it live on your own scenario and sample data, then explain your decisions, risks and impact.

Each team presents for 10 minutes and is scored out of 100.

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Choose Your Problem Statement

Each has the same structure: three user roles, five core expectations and three bonus ideas.

01

ENDING PLASTIC WASTE

Circular Plastics Exchange

Connect people who generate plastic waste with collectors and recyclers.

3 roles · 5 core · 3 bonus

02

SAFE PEOPLE AND PLANTS

SafeSite: A Digital Safety Companion

Make hazard reporting take under a minute and show where risk builds up.

3 roles · 5 core · 3 bonus

03

SUPPLY CHAIN AND LOGISTICS

Order-to-Delivery Control Tower

Track orders from plant to customer and predict delays before they happen.

3 roles · 5 core · 3 bonus

04

SMARTER BUSINESS PROCESSES

Smart Procure-to-Pay Assistant

Read supplier invoices and approve, flag or block them with clear reasons.

3 roles · 5 core · 3 bonus

Hypothetical scenario. Plant X is a fictional company.

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Circular Plastics Exchange

CONTEXT

Plant X, a fictional polymer manufacturer, wants more plastic waste to return to the recycling loop. Today, waste from housing societies, campuses and small businesses is mixed, unsorted and untraceable, so recyclers rarely receive the right material.

OBJECTIVE

Build a web or mobile platform that connects people and businesses who generate plastic waste with collectors and recyclers, and makes sorting, pickup and traceability simple.

WHAT WE ARE EXPECTING

1. Separate logins for waste generators, collectors and recyclers.
2. Generators upload a photo and the app identifies the plastic type (e.g. PET, HDPE, PP) with a confidence score.
3. Match each listing to a suitable collector or recycler by plastic type, quantity and location.
4. Schedule pickups and track every status, from requested to recycled.
5. Impact dashboard: kilograms diverted by plastic type and area, with an estimate of CO_2 avoided.

BONUS Rewards or points for generators · QR label on each batch for chain of custody ·

Hypothetical scenario. Plant X is a fictional company.

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SafeSite: A Digital Safety Companion

CONTEXT

At Plant X, a fictional chemical site, near-misses and hazards often go unreported because reporting is slow, paper-based or feels pointless. Contractors and new joiners are the most at risk and the least familiar with site rules.

OBJECTIVE

Build a mobile-first app that lets a worker report a hazard in under a minute, routes each report to the right person, and shows supervisors where risk is building up.

WHAT WE ARE EXPECTING

1. Separate logins for workers or contractors, supervisors and safety managers.
2. Report a hazard or near-miss with a photo, voice-to-text and location.
3. Automatically classify each report by hazard type and severity, and flag likely duplicates.
4. Turn every report into an assigned action, tracked to closure with due dates and escalation.
5. Dashboard with a heatmap by area and shift, trends and repeat hazards.

BONUS Offline reporting that syncs later · Contractor induction quiz before check-in · PPE check from a photo

Hypothetical scenario. Plant X is a fictional company.

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Order-to-Delivery Control Tower

CONTEXT

Plant X, a fictional polymer producer, ships product in bulk to business customers. An order passes through planning, dispatch, transport and delivery, yet customers and sales teams often learn an order is late only after it already is.

OBJECTIVE

Build a web app that tracks orders from plant to customer, predicts delays, and gives every stakeholder a clear, shared view.

WHAT WE ARE EXPECTING

1. Separate logins for the customer, the logistics coordinator and the transporter or driver.
2. Track orders and shipments on a map, with status milestones.
3. Predict an ETA and delay risk for each shipment, and show the reason.
4. Alert the customer and coordinator on delay risk, with a suggested action such as reroute or split shipment.
5. Dashboard of on-time performance and delay causes by route and transporter.

BONUS Truck slot booking at the plant gate · “Where is my order?” chatbot · Upload and download of invoice and proof of delivery

Hypothetical scenario. Plant X is a fictional company.

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Smart Procure-to-Pay Assistant

CONTEXT

The fictional finance team at Plant X processes large volumes of supplier invoices. Mismatches with purchase orders or goods receipts, duplicates and missing approvals cause late payments, delays and audit risk.

OBJECTIVE

Build a web app that reads supplier invoices, checks them against purchase records, and approves, flags or blocks each one with a clear reason.

WHAT WE ARE EXPECTING

1. Separate logins for suppliers, accounts payable clerks and approvers.
2. Suppliers upload invoices as PDF or image; the app extracts vendor, PO number, items, quantities, prices and tax.
3. Match invoice, PO and goods receipt; approve, flag or block with a reason (price, quantity, missing receipt).
4. Approval workflow with role-based thresholds and an audit trail of every decision.
5. Exceptions dashboard by type, supplier and age, and a supplier view of invoice status.

BONUS Duplicate-invoice detection · “Where is my payment?” chatbot · Simple controls report for auditors

Hypothetical scenario. Plant X is a fictional company.

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Five Ground Rules

Simple rules so every team competes on the same terms.

1

One statement per team

Choose one problem statement and cover all five core expectations. Bonus items are optional.

2

Build with any stack

Web or mobile, any language or framework. Open-source tools and APIs are welcome.

3

Create your own sample data

Fictional data only. No real, personal or confidential data, and none is supplied to you.

4

Run your own demo

Judges give no inputs or scenarios. You choose and run your own end-to-end flow live.

5

Generative AI is allowed. Declare it.

State where and how you used it. Undeclared use costs 10 points. You must be able to explain every part of your code, data and design.

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Your 10 Minutes

Each team gets one 10-minute session with the judges.

PROBLEM & APPROACH ~2 min	LIVE DEMO ~4 min	DEFEND ~2 min	JUDGES' Q&A ~2 min
Users, problem, key decision, where AI is used and where it is not	Your scenario run end to end on your own sample data	Controls, risks, assumptions, pilot or next step	At least one technical and one assumption question

HAVE THESE READY

A measurable impact

At least one indicator, with the calculation and the assumptions behind it.

A simple architecture

A diagram of how your solution is built, matching what you demo.

Your controls

How you handle AI risk: human review, confidence, privacy and access.

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How Teams Will Be Scored

Ten criteria, 100 points in total. The next slides show what each score level looks like.



Working prototype carries the most weight (20), then impact (15).

TOTAL 100

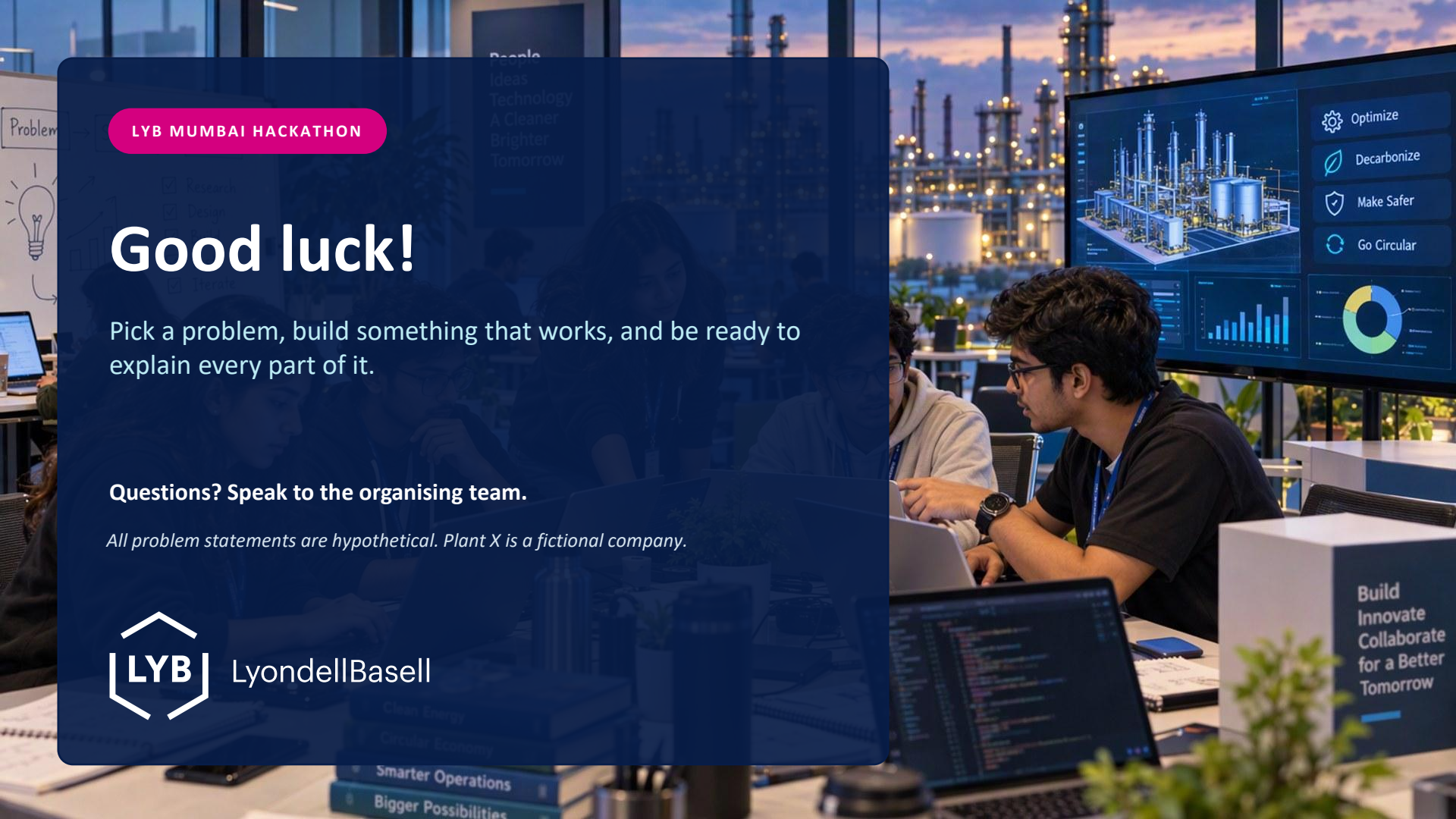
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Final Checklist

Run through this with your team before your session.

- [x] One statement chosen, all five core expectations covered
- [x] Your focus user and problem named
- [x] Working prototype ready to run live
- [x] Own sample data created, nothing real or confidential
- [x] Impact indicator with calculation and assumptions
- [x] Architecture diagram that matches your demo
- [x] Human review, confidence and privacy controls built in
- [x] Generative AI use declared
- [x] Every member can explain code, data and trade-offs

Build. Demo. Defend.


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Good luck!
Pick a problem, build something that works, and be ready to explain every part of it.
Questions? Speak to the organising team.
All problem statements are hypothetical. Plant X is a fictional company.
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Optimize
Decarbonize
Make Safer
Go Circular
Build Innovate Collaborate for a Better Tomorrow
Smarter Operations
Bigger Possibilities